L7. Explanation and Causes

Natural law

What it is?

A **natural law** is a scientific rule that describes a **fundamental and universal pattern in nature**.

It is usually very well-confirmed, applies everywhere, and does not depend on human decisions or culture.

Natural laws are stable, simple, and always true under the same conditions.

Example(s):

- **Newton's law of gravity:** Every mass attracts every other mass.
- **Ohm's law:** Voltage = Current × Resistance.
- **Law of conservation of energy:** Energy cannot be created or destroyed.
- **Boyle's law:** Pressure and volume of a gas are inversely related.

Law-like generalization

What it is?

A **law-like generalization** is a pattern we observe in data that *looks like* a natural law, but is **not guaranteed to be fundamental or universal**.

It is based on empirical observations and might hold most of the time, but there can be exceptions or limits.

Think of it as:

> "This seems to happen regularly in nature, based on what we've seen so far."

Example(s):

- **Bigger animals tend to live longer than smaller ones.** (Often true, but not always.)
- **High-pressure weather systems usually bring clear skies.**
- **In chromatography, larger molecules come out of the column first.**
- **People who exercise regularly tend to live longer.** (A strong pattern, but not a universal law.)

Deductive-Nomological account of explanation

What it is?

It is an idea that says we explain something by **using a scientific law plus facts about the situation** to **logically show why the event had to happen**.

In other words:

Law + starting facts = explanation

Example(s):

- **Why did the ball fall?**
  Law: Gravity pulls objects down.
  Fact: You dropped the ball.
  Conclusion: The ball must fall.
  This is a DN explanation.
- **Why is the shadow a certain length?**
  Law: Light creates shadows based on angles.
  Fact: The flagpole is a certain height.
  Conclusion: The shadow must have this length.

Why something can satisfy DN conditions but still not be a real explanation

Reason:

DN only checks if the event can be *logically deduced* from a law and some facts.

It does **not** check whether the explanation points to the **real cause**.

So DN accepts explanations that work on paper but make no sense in reality.

Examples

1. Flagpole and shadow

You can deduce the pole height from the shadow.

DN says it is a valid explanation.

But the shadow does **not** cause the pole's height.

So it is not a real explanation.

2. Mr. Jones cannot get pregnant

DN uses the rule "men who take birth-control pills don't get pregnant."

It is logically correct, but the pills are irrelevant.

The real cause is being male.

3. Pendulum period and length

DN says either variable can explain the other.

But only the **length causes the period**, not the other way around.

Explanandum

What it is?

The **explanandum** is the thing you want to explain.

It is the **question or event** you are asking about.

Example(s):

- A vase broke.
  → This is the explanandum.
- Why did the car crash?
  → The car crash is the explanandum.
- Why is the sky blue?
  → The blueness of the sky is the explanandum.

Explanans (explanantia)

What it is?

The **explanans** is the information that **gives the explanation**.

It is what you use to explain the thing you are asking about.

Example(s):

- Explanandum: The vase broke
  Explanans: Someone bumped the table and the vase fell.
- Explanandum: The car overheated
  Explanans: The radiator was empty and the engine got too hot.
- Explanandum: The sky is blue
  Explanans: Light scatters in the atmosphere.

Singular causal explanation

What it is?

A **singular causal explanation** explains one specific event by listing the **specific things that caused it**.

It does not use general laws. It uses **particular events** that happened in this case.

Example(s):

- Why did the vase break?

  Because

  1. It was on the edge of the table
  2. You bumped the table
  3. The vase fell and hit the floor

     → These specific events caused the break.
- Why did the car stop?

  Because

  1. The fuel tank was empty
  2. The engine could not get fuel

     → These particular events explain this one situation.

Singular explanandum

What it is?

A **singular explanandum** is one specific event or thing that you want to explain.

Example(s):

- Why did **this vase** break?
- Why did **this car** crash?
- Why is **this person** sick?

General explanandum

What it is?

A **general explanandum** is a general pattern or trend you want to explain.

It is not about one event but about many cases.

Example(s):

- Why do **penguins** live in cold places?
- Why does **inflation** rise when interest rates drop?
- Why do **heavier cars** use more fuel?

Singular explanans

What it is?

A **singular explanans** explains something using **specific events** that happened in this particular case.

Example(s):

- The vase broke because
  1. You hit the table
  2. The vase fell
  3. It hit the floor
- The car stopped because
  1. The fuel tank was empty
  2. The engine lost power

General explanans

What it is?

A **general explanans** explains something by using a **general rule or law**, not specific events.

Example(s):

- Objects fall because of **gravity**.
- Gases expand when heated because of the **gas laws**.
- Larger animals live longer because of **biological scaling laws**.

Contrastive explanandum

What it is?

A **contrastive explanandum** is a question that asks

"Why this instead of that?"

It means you are not just asking why something happened, but why it happened **in this way rather than another possible way**.

Example(s):

- Why did **this car** crash **here** instead of **somewhere else**?
- Why did the vase break **now** instead of **later**?
- Why did the storm hit **Stockholm** instead of **Uppsala**?
- Why did this person choose **A** instead of **B**?

Contrastive explanans

What it is?

A **contrastive explanans** gives the explanation that answers a **contrastive question**.

It explains **why one outcome happened instead of another** by pointing to the **specific difference-making cause**.

Example(s):

- Why did the car crash **here** instead of **down the road**?

  Contrastive explanans: This curve was **tighter**, while the other part of the road was **straight**.
- Why did the storm hit **Stockholm** instead of **Uppsala**?

  Contrastive explanans: The **pressure system** moved east, not north.
- Why did this student pass **instead of** failing?

  Contrastive explanans: They **studied extra** the last week, which made the difference.

Accuracy

What it is?

Accuracy means the explanation must describe the **real world correctly**.

The explanans must mention the **actual cause** that truly made the difference.

It should not include false information or irrelevant causes.

In short:

A good explanation must point to the real difference-making cause.

Example(s):

- The vase broke because the **cat pushed it off the table**.

  (Accurate, because this is what really happened.)
- The vase broke because of **a ghost**.

(Not accurate, because it does not describe the actual cause.)

- The car skidded because the **road was icy**, not because of the color of the car.

  (The icy road is the real difference-making cause; car color is irrelevant.)

Precision (of the explanandum)

What it is?

Precision means the question you are trying to explain must be **clear and specific**.

You must clearly state **what the contrast is**, meaning:

What exactly are we comparing?

Why this instead of what?

A precise explanandum makes it easier to give a good explanation.

Example(s):

- Vague:

  "Why did the car crash?"

  (Not precise, many possible answers.)
- Precise:

  "Why did the car crash **here** instead of **around the corner**?"

  (Clear contrast.)
- Vague:

  "Why did the vase break?"
- Precise:

  "Why did the vase break **today** instead of **yesterday**?"

Difference-Making (of the explanans)

What it is?

The explanans must point to the **specific cause(s)** that actually made the difference in the situation.

It should answer the contrast directly:

Why this outcome instead of the other one?

Only the **difference-making causes** matter.

Extra details that did not affect the outcome should not be included.

Example(s):

- Explanandum: Why did the car crash **here** instead of **further down the road**?

  Difference-making explanans:

  "The curve **here** was too sharp."

  (This is the cause that made the difference.)
- Explanandum: Why did this plant grow **faster** than the other plant?

  Explanans: The faster-growing plant got **more sunlight**.

  (This is the difference-making factor.)
- Explanandum: Why did student A pass **instead of** failing?

  Explanans: They **studied more** during the last week.

  (This directly produced the difference.)

Non-sensitivity (of the explanans)

What it is?

Non-sensitivity means the explanation should **not depend on a long chain of fragile background conditions**.

A good explanans works **even if small details in the background change**.

If the explanation depends on many things all being exactly right, then it is **too sensitive** and not reliable.

In short:

A good explanation should not fall apart if one tiny background detail changes.

Example(s):

- Bad (too sensitive):

  "The revolution happened because Lenin skipped listening to Beethoven that day."

  (This depends on many unrealistic background assumptions.)
- Good (non-sensitive):

  "The revolution happened because of political unrest and economic collapse."

  (These causes do not depend on fragile background conditions.)

Cognitive salience

What it is?

Cognitive salience means the explanation should be **easy to understand** for the person who is asking the question.

A good explanation is clear, simple, and uses ideas the audience can easily grasp.

In short:

The easier the explanation is to understand, the better it is.

Example(s):

- Easy to grasp (good):

  "The car skidded because the road was icy."

  (Simple and clear.)

- Hard to grasp (not good):

  "The car skidded due to complex friction coefficients and micro-level tire interactions."

  (True but too complicated for most people.)
- Easy to grasp (good):

  "The plant died because it got no water."
- Hard to grasp (not good):

  "The plant died because of a drop in cellular turgor pressure caused by osmotic imbalance."

  (Scientifically correct but not cognitively simple.)

Direct cause

What it is?

X is a **direct cause** of Y if changing X would **immediately change Y**, while everything else stays the same.

In other words:

If you can **intervene on X** and Y changes because of that, then X directly causes Y.

Example(s):

- Turning the **light switch** (X) directly causes the **light to turn on** (Y).

  If you flip the switch, the light changes, even if everything else stays the same.
- Pressing the **brake pedal** (X) directly causes the **car to slow down** (Y).

  Changing X changes Y.
- Heating **water** (X) directly causes the **temperature** (Y) to rise.

  If you add heat, the water gets hotter.

Contributing cause

What it is?

A **contributing cause** is something that helps bring about an outcome **through a chain of direct causes**.

X is a contributing cause of Y if:

- X starts a **causal chain**,
- each step in the chain is a **direct cause**, and
- the chain eventually ends at Y.

X might not directly cause Y, but it still **contributes** to Y happening.

Example(s):

- You **push a domino** (X).

  That domino hits another, and eventually the last one **falls** (Y).

  → X is a contributing cause of Y through the chain.
- You **light a match** (X).

  The flame lights a candle, which heats wax, which melts it (Y).

  → Your action contributed to Y through multiple steps.
- A **traffic jam** (X) slows cars behind it, which delays people, which causes someone to **arrive late** (Y).

  → X contributed to Y through a series of direct effects.

Correlation

What it is?

Correlation means that two variables vary together in a systematic way. When one variable changes the other tends to change as well. Correlation is symmetric. If X is correlated with Y then Y is also correlated with X. Correlation by itself does not show that one variable causes the other.

Example(s)

- Ice cream sales and temperature are correlated
- Shoe size and reading ability are correlated in children because age affects both
- Exercise levels and health outcomes are correlated
- Screen time and sleep quality are correlated